

Gustav Vallinder

Location

Stockholm, Sweden

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About

Control engineer and software developer, currently pursuing an industrial PhD in automatic control for autonomous driving at Traton and KTH. I enjoy combining algorithms, physics and software to make things move. After finishing my PhD, I intend to continue my career in industrial R&D.

Experience

Industrial PhD Student

Traton AB, KTH Royal Institute of Technology | 2022.10 - Present

- Conducting motion control research in the AllDrive project to enable safe autonomous driving of semi-trailer trucks in winter conditions.
- Developing friction-aware motion control algorithms and software for testing at winter proving grounds in northern Sweden.
- Collaborating cross-functionally with students, researchers and engineers in industry and academia.

Software Developer

Epiroc Rock Drills AB | 2021.08 - 2022.08

- Developed control, obstacle avoidance and localization features for autonomous underground mining trucks and LHDs.
- Improved the performance of vehicles at customer sites through tuning, in-vehicle tests and analysis of log data.

Education

Master of Science in Electrical Engineering

Uppsala University | 2016.08 - 2021.06

- Specialization on systems and control with courses in control theory, signal processing, embedded systems, machine learning, and optimization.
- Spent a year as an exchange student in Germany at the Technical University of Munich.
- Did master thesis with Epiroc to improve speed and accuracy of autonomous LHDs using path following model predictive control.

Skills

Interests

Autonomous Systems, Vehicle Dynamics, Model Predictive Control, Modelling and Simulation, State Estimation.

Software

C++, Python, MATLAB, Linux, Git, Jira.

Publications

G. Vallinder, S. Bhat and J. Mårtensson, "Time-Optimal Lane Change for a Tractor-Semitrailer on Varying Road Friction: Performance Bounds for Autonomous Driving", *12th IFAC Symposium on Intelligent Autonomous Vehicles IAV 2025*, Phoenix, AZ, United States, 2025.

G. Vallinder, J. Mårtensson, P. F. Lima and S. Bhat, "Model Predictive Control for Autonomous Driving: Comparing Kinematic and Dynamic Models of Tractor-Trailer Systems", *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024.

Additional Experience

Teaching

Automatic Control, Basic Course, KTH Royal Institute of Technology, 2023.

Digital Technology and Electronics, Uppsala University, 2019.

Digital Electronics, Uppsala University, 2018.

Volunteering

Head of Corporate Relations, Uppsala Studentförening Elektroteknik, 2018.

Student Union Council Rep., Uppsala Studentförening Elektroteknik, 2017.

Merits

Memberships

Affiliated PhD Student, WASP Graduate School.

Affiliate, IFAC International Federation of Automatic Control.

Awards

Forsmark Kraftgrupp AB Scholarship, 2015.